Healthcare Innovation Challenge

2026 Challenge Submission Form

Name:	Sarah Monahan	Contact No.:	01 4142790
Job Title:	Discharge planning ocupational therapist	Directorate:	Jerome Powell, COO
Idea Title:	Smarter equipment tracking for better patient care		

Challenge Description:

Describe what the problem is you are trying to solve. This should include a statement of the current process, problem and what are the challenges, shortcomings etc.

Current Process
The occupational Therapy department maintains a stock of specialist seating and manual handling eqyuipment, including specialised and manual wheelchairs, pressure-relieving cushions, elevating leg rest, headrests, residual limb boards and sliding boards. This equipment is stored within the Occupation Therapy Deparmtnet and is issued to inpatients based on clinical need.
Currently equipment is tracked using an excel based 'Acute Equipment tracker' stored within a shared folder. Each equipment category has a separate tab containing information such as the asset number, barcode, serial number and service information.
When eqipment is issued to a patient, the prescribing Occupational Therapist is responsible for manually updating the tracker with details including:
Patient Name and medical records number; Equipment issued; Location of the equipment; Date issued; Whether the equipment is onl load.
The tracker also uses colour coding to indicate whether equipment requires repair / service, has been discharged with a patient on short-term loan, or is missing.
There is also a "waitlist" tab, where clinicians record equipment required for inpatients. When an item becomes available, staff are expected to check the waitlist and allocate the equipment accordingly.
An additional pool of equipment funded through the Winter Initiative and managed by the Discharge Planning Occupational Therapy service is also tracked manually using an excel spreadsheet.

Problems, challenges and short comings:

The current process is highly dependent on manual data entry and staff compliance. This creates a number of challenges:

1) Time-consuming administrative process: Manually recording, updating and checking equipment is lenghty and efficient. This administrative burden reduces the time available for direct clinical contact with patients.
2) The time required for stock checks: Therapy assistants undertake montly stock checks to confirm that equipment is physically located where the tracker indicates. This is a lengthy process and reuqires staff time that could otherwise be directed towards patient care.
3) Delayed or retrospective updates: Equipment information is frequently entered retrospectively rather than at the point of issue or return. As a result, the tracker may not provide an accurate real-time representation of equipment availability.
4) Reliance on manual compliance: The effectiveness of the current system relies on multiple staff members

consistently remembering to update the tracker. This creates an inherent risk of information becoming incomplete or outdated.

5) Equipment going missing: Equipjment has been lost or cannot be located, resulting in financial costs associated with replacing or reimbursing equipment. This also reduces the pool of equipment available for partients.
6) Impact on patient access: Reduced requipment availabitlity can result in delays in providing patients with appropriate specialist seating, pressure care and postural managmeent equipment. This has the potential to impact patient comfort, safety and timely discharge.
7) Waitlist management: The current waitlist process is difficult to manage efficiently. Staff must log into the shared folder, locate the relevant spreadsheet and review the waitlist before allocating available equipment. Due to the time required to access and check the information, equipment may be allocated without the waitlist being reviewed. This can result in equipment not being allocated according to the existing order of clinical need or waiting time.
8) Difficulty identifying equipment status. A large volume of information is contained within the spreadsheet, makinig it difficult ot quickly determine which equipment is available, on loan, awaiting repair, missing or allocated to another patient.
9) Limited real-time visibility: There is currently no simple, centralised way for the wider Occupation Therapy team to see the real-time status and location of equipment.

Overall, the current system is functional but relies heavily on manual processes and individual compliance. This creates avoidable administrative workload, reduces visibility of available equipment, increases the rish of equipment being lost or unavailable, may contribute to delays in providing appropriate equipment to patients.
The Occupational Therapy Department currently manages approximately 155 products of equipment across nine equipment categories, making manual tracking increasingly difficult as the equipment pool grows.

Idea Goal:

Describe how solving the current would improve the current situation, overall impact within you area, applicability to the wider hospital.

A more effective tracking system could:

1) Improve timely access to equipment: Staff would be able to quickly identify where equipment is located and wheterh it is available, reducing delays in providing patients with appropriate specialist seating, pressure care and postural management equipment.
2) Improve patient care and experience: Timely access to appropriate equipment can support patient comfort, positioning, pressure care, mobility and safe discharge planning.
3) Reduce administrative workload: Automating or simplifying the process of issuing, returning and locating equipment would reduce the amount of time Occupational Therapists and Therapy Assistants spend manually updating spreadsheets and completing physical stock checks. This time could instead by redirected towards direct patient care and clinical activity.
4) Improve staff experience: A more efficient system would reduce the frustration and time associated with searching for equipmnet, checking multiple records and manually reconciling equipment locations.
5) Improve equpiment utilisation: A real-time overview of equipment availability would allow the existing equipment pool to be used more effectively. Staff could quickly identify equipment that has become available and alloate it to the next appropriate patient rather than purchasing or sourcing additional equipment unnecessarily. We already own equipment, but because we cannot reliably identify where it is or whether it is available, we cannot maximise the use of the equipment we already have.
6) Improve management of the equipment waitlist: A centralised system could make outstanding equipment requests more visible and support a more consistent process for allocating equipment when it becomes available.
7) Reduce equpment loss and associated costs: Over the past three years, approximately €4,600 per annum has been incurred in relation to equipment that has gone missing and could not be located within Tallaght university Hospital. Improving equipment visibility and accountability has the potential to reduce these avoidable costs.
8) Improve governance and accountability: A more structed tracking system would provide a clearer record of when equipment was issued, to who it was issued, where it is located an when it was returned. This would improve oversight of the equipment inventory and support more accurate asset management.
9) Potential wider hospital applicability: Although the initial focus would be on Occupation Therapy equipment, the principle could potentially be applied to other hospital departments that manage shared

clinical equipment, particularly where equipmnet is frequently loaned, transferred between locations or difficult to track.

Ultimately, the aim is to move from a predominantly manual and restrospective process to a more efficient system that provides accurate, accessible and time information about equipment throughout its lifecycle.

Location:

What areas within the Hospital does this apply to?

The proposed improvement would initially apply to the Occupation Therapy Department and all inpatients wards within Tallaght University Hospital.

Signed: Sarah Monahan **Date:** 21/09/2026